
Reference

- T.Y.Lam
- Noncommutative Noetherian Ring, J.C.McConnell

Chapter 1 Free Modules, Projective and Injective Modules

1.1 Free Modules

1.1.1 IBN

Category of Left/Right Modules

Definition 1.1.1.1

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Free Module

◇ Lemma 1.1.1.2 (Generation Lemma)

◦ Proof ...

□

★ Corollary 1.1.1.3

◦ Proof ...

□

IBN Properties

Definition 1.1.1.4 ...

1.1.2 Stable Finiteness

Definition 1.1.2.1 Recall: A ring is **Dedekind-finite** if

$$c, d \in S, cd = 1 \Rightarrow dc = 1$$

A ring R is **stably finite** if the matrix $M_n(R)$ are Dedekind-finite.

Remark Dedekind finiteness: In $M_n(F)$, we have

$$AB = I \Rightarrow BA = I$$

With:

...

But for $M_\omega(F)$, a counterexample is

...

♠ **Proposition 1.1.2.2** The following properties of R are equivalent:

- R is stably finite;
- For any n , $R^n \cong R^n \oplus N \Rightarrow 0$ in $\mathfrak{m}_R$;
- For any n , any epimorphism $R^n \rightarrow R^n$ in $\mathfrak{m}_R$ is an isomorphism.

◦ **Proof** ... □

♠ **Proposition 1.1.2.3** For any nonzero element R , stable finiteness implies IBN. But not conversely.

◦ **Proof** ... □

♠ **Proposition 1.1.2.4** ...

◦ **Proof** ... □

1.1.3 The Rank Condition

Definition 1.1.3.1

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♠ **Proposition 1.1.3.2** ...

◦ **Proof** ... □

1.1.4 The Strong Rank Condition

1.2 Projective Modules

1.2.1 Definitions, Examples

Definition 1.2.1.1 ...

1.3 Injective Modules

Chapter 2 Flat Modules and Homological Dimension

2.1 Flat and Faithfully Flat Modules

Definition 2.1.0.1 ...

Chapter 3 Jacobson Radical of Non-commutative Rings

Definition 3.0.0.1 (Jacobson Radical) Define: The Jacobson radical

$$\text{rad}(R) = \cap_{I \in L} I \quad L = \{I \subseteq R \mid I \text{ is the left ideal of } R\}$$

◇ **Lemma 3.0.0.2** For $y \in R$, the following statements are equivalent:

- $y \in \text{rad}(R)$;
- $1 - xy$ is left-invertible for any $x \in R$;
- $yM = 0$ for any simple left R -module M .

◦ **Proof**

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□

Annihilators

Definition 3.0.0.3 ...

Jacobson Semisimple

Nilpotent

3.1 Jacobson Radical Under Change of Rings

3.2 Group Ring and the J-Semisimplicity

Chapter 4 Division Algebra

Reference

- A First Course in Noncommutative Rings, T.Y.Lam, Chapter 5;
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4.1 Finite Dimensional Division Algebra

4.1.1 Discussion of Complex Numbers, Quaternion and Frobenius's Theorem

Introduction We have the extension of field

$$\mathbb{R} \rightsquigarrow \mathbb{C} \quad \dim_{\mathbb{R}}(\mathbb{C}) = 2$$

Moreover, we want to find some $\mathbb{R}$ -algebras, and be able to divide numbers.

Definition 4.1.1.1 A unital algebra D in which every nonzero element is invertible is called a division algebra.

i.e. for every pair $a \in D - \{0\}$, such x, y exists.

$$ax = 1 \quad ya = 1$$

A question is:

Is it possible to define multiplication on an n -dimensional real space so that it becomes a real division algebra?

Standard Results:

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Assume: D is an n -dimensional $\mathbb{R}$ -algebra.

◇ **Lemma 4.1.1.2** For every $x \in D$, there exists a $\beta \in \mathbb{R}$ such that

$$x^2 + \beta x \in \mathbb{R}$$

◦ **Proof** ...

□

Define:

$$V := \{v \in D \mid v^2 \leq 0\}$$

◇ **Lemma 4.1.1.3** If $u, v \in V$ are linearly independent, then so are u, v and 1.

◦ **Proof** ...

□

Jordan Product

Definition 4.1.1.4 (Jordan Product) For $x, y \in D$, we write

$$x \circ y := xy + yx$$

Frobenius's Theorem

► **Theorem** (Frobenius's Theorem) A **finite dimensional division algebra** D is isomorphic to $\mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

4.2 Structure of Finite Dimensional Algebras

Chapter 5 Wedderburn-Artin-Theory

5.1 Simple, Semisimple Modules

Definition 5.1.0.1

- A nonzero module M is **simple** or **irreducible** if it contains no proper non-zero submodule.
- A module M is called **semisimple** or **completely module** R -module if M is a direct sum of simple modules.

Definition 5.1.0.2

An R -module is **cyclic** with generator $m \in M$ such that

$$M = Rm$$

5.1.1 Simple Modules

Definition 5.1.1.1

An **irreducible representation** of an algebra A is defined as a representation

$$\varphi : A \rightarrow \text{End}_F(V)$$

which contains no proper nonzero subspace W of V that is invariant under $\varphi(a), a \in A$.

Equivalently, the associated module with structure $A \times V \rightarrow V, (a, v) \mapsto \rho(a)(v)$. The module V is irreducible.

♠ **Proposition 5.1.1.2** The following are equivalent for R -module M :

- M is simple;
- M is cyclic and every nonzero element is a generator;
- $M \cong R/I$ for some maximal left ideal I ;

◦ Proof

- (1) $\Rightarrow$ (2) M is simple. Then for any $m \in M$, either $Rm = 0$ or $Rm = M$. Take

$$N := \{m \in M \mid Rm = 0\}$$

Then, N is a submodule, $N = 0$. This implies, that for every $m \neq 0, Rm = M$;

- (2) $\Rightarrow$ (3) Take

$$\phi : R \rightarrow Rm \quad r \mapsto rm$$

And $I := \text{Ker}(\phi)$. I is a submodule of R , and $R/I \cong Rm = M$.

I must be maximal. If not,

- (3) $\Rightarrow$ (1) R/I contains no submodule.

□

♣ **Example 5.1.1.3** Let D be a division ring and V is vector space over D . Then D is simple $\Leftrightarrow \dim_D(V) = 1$.

♣ **Example 5.1.1.4** Let V be a nonzero vector space over division ring D .

V can be regarded as a module over $R = \text{End}_D(V)$, via $fv := f(v)$. We claim that V is a simple R -module: For each pair $v, w \in V$ with $v \neq 0$, there exist some f such that $f(v) = w$. This is true.

♣ **Example 5.1.1.5** Let R be a ring and L be an ideal. Then L is simple L -module iff L is minimal. Omitted.

Minimal Left Ideals

◇ **Lemma 5.1.1.6** Let R be a simple ring having a minimal left ideal L . Then every simple R -module is isomorphic to L .

◦ **Proof** ...

□

Maximal Left Ideal

Definition 5.1.1.7 A left ideal U of a ring R is called a **maximal left ideal** if $U \neq R$ and U is not properly contained in a proper left ideal.

♣ **Example 5.1.1.8** Let $R = M_n(D)$, with D a division ring. And let

$$U = RE_{11} \oplus \cdots \oplus RE_{n-1, n-1}$$

It is easy to check: U is a maximal left ideal of R .

And:

$$R = U \oplus L$$

where $L = RE_{nn}$ is the minimal left ideal. Especially we have $R/U \cong L$.

◇ **Lemma 5.1.1.9** Let R be a unital ring. Then every proper left ideal L of R is contained in a maximal left ideal.

◦ **Proof** Let $\mathfrak{S}$ be the set of all proper left ideals of R contains L ,

$$S = \{I \mid I \text{ is left ideal and } L \subseteq I\}$$

Now, let $\{K_i \mid i \in I\}$ be a chain in S . Then

$$K = \cup K_i$$

is its upper bound. K is a left ideal, $1 \notin K \Rightarrow K \neq R$. By Zorn's lemma,

□

◇ **Lemma 5.1.1.10** Let R be a ring. Every simple R -module is isomorphic to R/U for some maximal left ideal U of R .

Conversely, if U is a maximal ideal of R such that $R^2 \not\subseteq U$. Then R/U is simple R -module.

◦ **Proof** ...

□

★ **Corollary 5.1.1.11** Every nonzero unital ring has a simple module.

◦ **Proof** ...

□

5.1.2 Semisimple Modules

♣ **Example 5.1.2.1** ...

◇ **Lemma 5.1.2.2** If M is the sum of a family of simple submodules $\{M_i | i \in I\}$. Then, there exists a subset $K \subseteq I$ such that

$$M = \bigoplus_{k \in K} M_k$$

◦ **Proof** ...

□

★ **Corollary 5.1.2.3** If a module is the sum of a family of simple submodules, then it is semisimple.

◦ **Proof** Omitted.

□

◇ **Lemma 5.1.2.4** The following statements are equivalent for a nonzero module M :

- M is semisimple;
- Every submodule of M is a direct summand and $Rm \neq 0$ for every nonzero $m \in M$;

◦ **Proof** ...

□

★ **Corollary 5.1.2.5** A **unital** module M is semisimple iff every submodule of M is a direct summand.

◦ **Proof** Omitted.

□

Endomorphisms Rings of Semisimple Modules

► **Theorem** (Schur's Lemma) If M is a simple R -module. Then the endomorphism ring $End_R(M)$ is a division ring.

◦ **Proof** Let $f \in End_R(M)$ and $f \neq 0$. Then, $ker(f)$ is a submodule of M , $Ker(f) = 0$. Then, $f \in Aut_R(M)$. f is inversable. There exists some $f^{-1} \in Aut_R(M)$.

□

A slightly more general version:

- A nonzero homomorphism from an arbitrary module to a simple module is surjective, and;
- A nonzero homomorphism from a simple module to an arbitrary module is injective.

◇ **Lemma 5.1.2.6** If M, N are simple modules, when they are not isomorphic, there are no nonzero homomorphisms from one to another.

◦ **Proof** ...

□

Semisimple Rings

5.1.3 Chain Conditions

5.2 Wedderburn-Artin Theorem